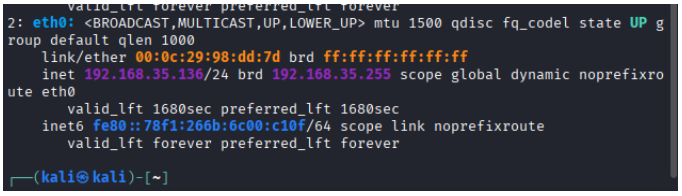
Lab01:BatGoiTin(mini-lab)

B1: Kiem tra IP
B2: Bat dich vu telnet tren Server và dat password administrator/123
B3: Cai Wireshark tren may attacker
B4: Client ket noi voi server thong qua dich vu telnet - Attacker bat goi tin va lay dc password

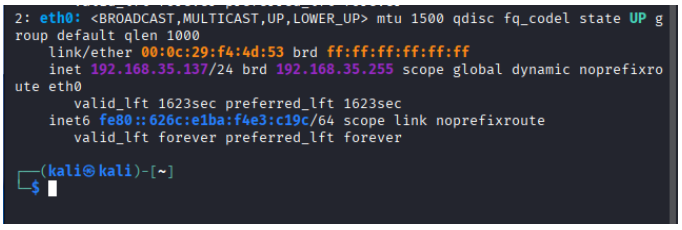
tạo thêm webserver để đăng nhập vào
lab này

Kiem tra IP

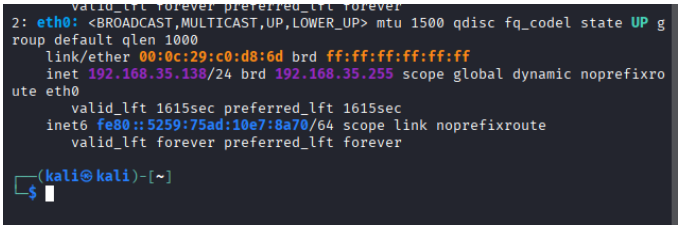
Server



Client



Attacker



Setup

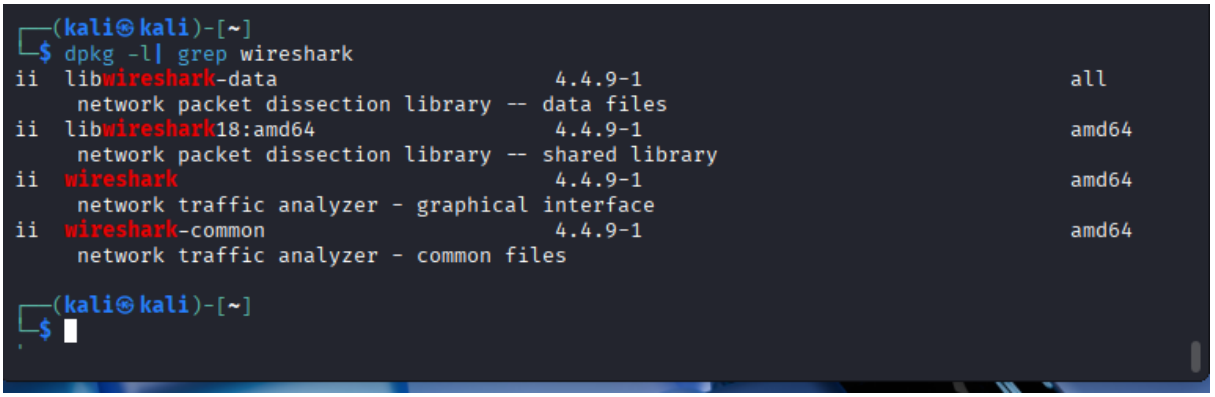
Server

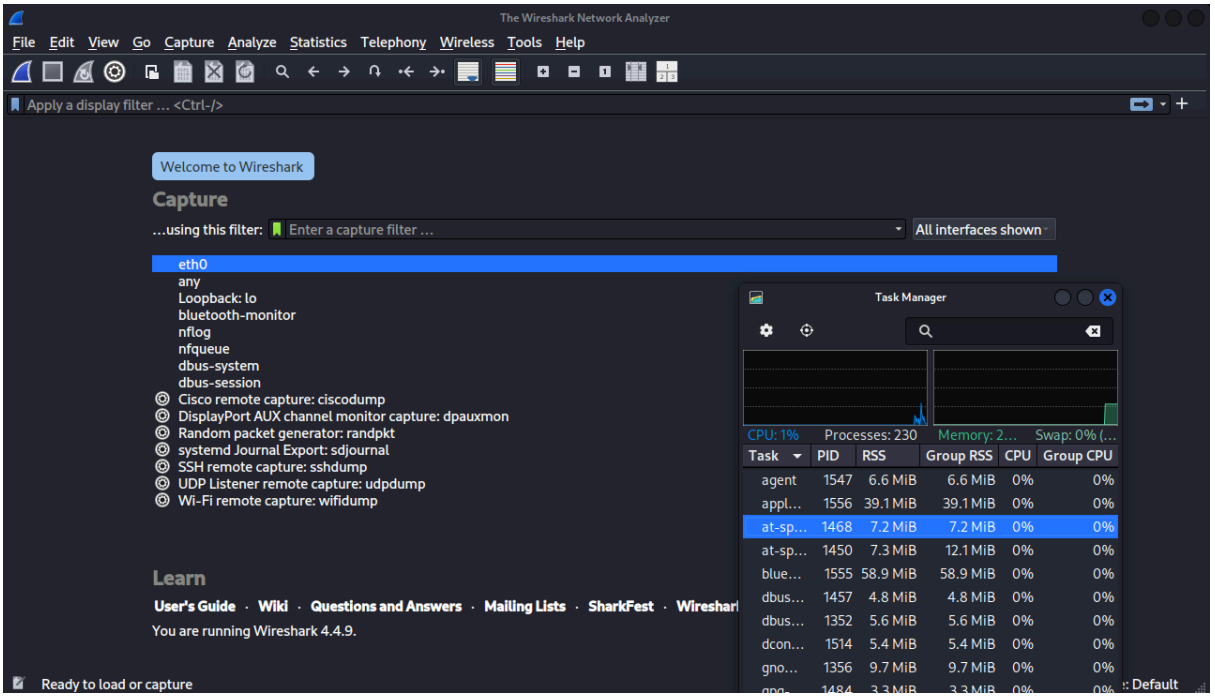
```
kali@kali: ~
Session  Actions  Edit  View  Help
ii  adwaita-icon-theme                    49.0-1                          >
ii  aircrack-ng                           1:1.7+git20230807.4bf83f1a-2+b1 >
ii  alsa-topology-conf                    1.2.5.1-3                       >
ii  alsa-ucm-conf                         1.2.14-1                        >
ii  amass                                 5.0.1-0kali2                    >
ii  amass-common                          4.2.0-0kali1                    >
ii  amd64-microcode                       3.20250311.1                    >
ii  apache2                               2.4.65-3+b1                     >
ii  apache2-bin                           2.4.65-3+b1                     >
ii  apache2-data                          2.4.65-3                        >
ii  apache2-utils                         2.4.65-3+b1                     >
ii  apparmor                              4.1.0-1                         >
ii  apt                                   3.1.8+kali1                     >
ii  apt-file                              3.3                             >
ii  apt-utils                             3.1.8+kali1                     >
ii  arj                                   3.10.22-28                      >
ii  arp-scan                              1.10.0-2+b1                     >
ii  arping                                2.26-1+b1                       >
lines 1-27
zsh: suspended  dpkg -l

┌──(kali㉿kali)-[~]
└─$ dpkg -l| grep net-tools
ii  net-tools                             2.10-2
       amd64        NET-3 networking toolkit

┌──(kali㉿kali)-[~]
└─$
```

Client

```
┌──(kali㉿kali)-[~]
└─$ dpkg -l| grep telnet
ii  inetutils-telnet                      2:2.6-4                         amd64
     telnet client
ii  libcli1.10t64:amd64                   1.10.7-2                        amd64
     emulates a cisco style telnet command-line interface
ii  ruby-net-telnet                       0.2.0-1                         all
     telnet client library
ii  telnet                                0.17+2.6-4                      all
     transitional dummy package for inetutils-telnet default switch

┌──(kali㉿kali)-[~]
└─$
```

Attacker





Install and update on Server

Install and config telnet

I configured the Telnet server using the `inetd` super-server daemon, which is a more traditional approach compared to using `systemd` services directly.

- sudo apt update
- sudo apt install openbsd-inetd telnetd -y
- which telnetd

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install openbsd-inetd telnetd -y
openbsd-inetd is already the newest version (0.20221205-3+b2).
telnetd is already the newest version (0.17+2.6-4).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 37

┌──(kali㉿kali)-[~]
└─$ which telnetd
/usr/sbin/telnetd

┌──(kali㉿kali)-[~]
└─$
```

- sudo adduser telnetuser

```
┌──(kali㉿kali)-[~]
└─$ sudo adduser telnetuser
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for telnetuser
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y

┌──(kali㉿kali)-[~]
└─$
```

- sudo nano /etc/inetd.conf

```
kali@kali: ~
Session  Actions  Edit  View  Help
  GNU nano 8.6                          /etc/inetd.conf
# /etc/inetd.conf: see inetd(8) for further informations.
#
# Internet superserver configuration database.
#
#
# Lines starting with "#:LABEL:" or "#<off>#" should not
# be changed unless you know what you are doing!
#
# If you want to disable an entry so it is not touched during
# package updates just comment it out with a single '#' character.
#
# Packages should modify this file by using update-inetd(8).
#
# <service_name> <sock_type> <proto> <flags> <user> <server_path> <args>
#
#:INTERNAL: Internal services
#discard                 stream  tcp6    nowait  root    internal
#discard                 dgram   udp6    wait    root    internal
#daytime                 stream  tcp6    nowait  root    internal
#time            stream  tcp6    nowait  root    internal

#:STANDARD: These are standard services.
telnet  stream  tcp     nowait  root    /usr/sbin/tcpd  /usr/sbin/telnetd

#:BSD: Shell, login, exec and talk are BSD protocols.

#:MAIL: Mail, news and uucp services.

#:INFO: Info services
                                        [ Read 39 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line
```

- sudo systemctl restart openbsd-inetd
- sudo systemctl enable openbsd-inetd
- sudo systemctl status openbsd-inetd

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status openbsd-inetd
● inetd.service - Internet superserver
     Loaded: loaded (/usr/lib/systemd/system/inetd.service; enabled; preset: disabled)
     Active: active (running) since Tue 2025-11-11 06:20:25 EST; 1min 1s ago
 Invocation: d9af816ad9794f3eab49e12fb3bd753b
       Docs: man:inetd(8)
   Main PID: 132010 (inetd)
      Tasks: 1 (limit: 4535)
     Memory: 508K (peak: 1.7M)
        CPU: 14ms
     CGroup: /system.slice/inetd.service
             └─132010 /usr/sbin/inetd

Nov 11 06:20:25 kali systemd[1]: Starting inetd.service - Internet superserver...
Nov 11 06:20:25 kali systemd[1]: Started inetd.service - Internet superserver.

┌──(kali㉿kali)-[~]
└─$
```

- sudo netstat -tulpn | grep :23

```
┌──(kali㉿kali)-[~]
└─$ sudo netstat -tulpn |grep :23
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN      132010/inetd

tcp6       0      0 :::23                   :::*                    LISTEN      132010/inetd


┌──(kali㉿kali)-[~]
└─$
```

On Client

- telnet 192.168.35.136

```
┌──(kali㉿kali)-[~]
└─$ telnet 192.168.35.136 23
Trying 192.168.35.136 ...
Connected to 192.168.35.136.
Escape character is '^]'.

Linux 6.12.38+kali-amd64 (kali) (pts/1)

kali login: telnetuser
Password:
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
┌──(telnetuser㉿kali)-[~]
└─$
```

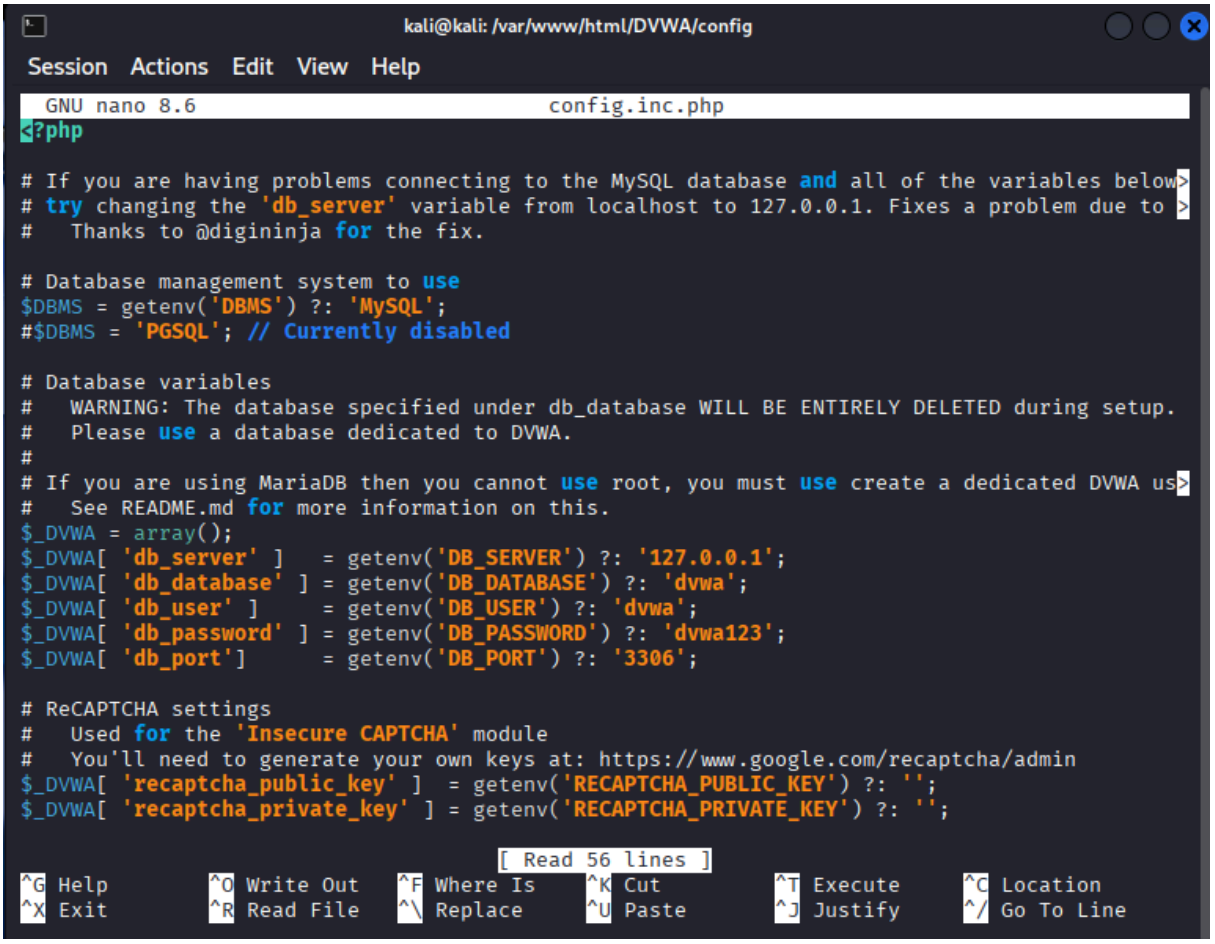
Install and config HTTP with DVWA

- sudo apt install apache2 php php-mysql libapache2-mod-php mariadb-server -y

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install apache2 php php-mysql libapache2-mod-php mariadb-server -y
apache2 is already the newest version (2.4.65-3+b1).
php is already the newest version (2:8.4+96).
php-mysql is already the newest version (2:8.4+96).
libapache2-mod-php is already the newest version (2:8.4+96).
libapache2-mod-php set to manually installed.
mariadb-server is already the newest version (1:11.8.3-1+b1).
mariadb-server set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 37

┌──(kali㉿kali)-[~]
└─$
```

- sudo chown-R www-data:www-data/var/www/html/DVWA
- sudo chmod-R755/var/www/html/DVWA

- cd /var/www/html/DVWA/config
- sudo cpconfig.inc.php.dist config.inc.php
- sudo nanoconfig.inc.php



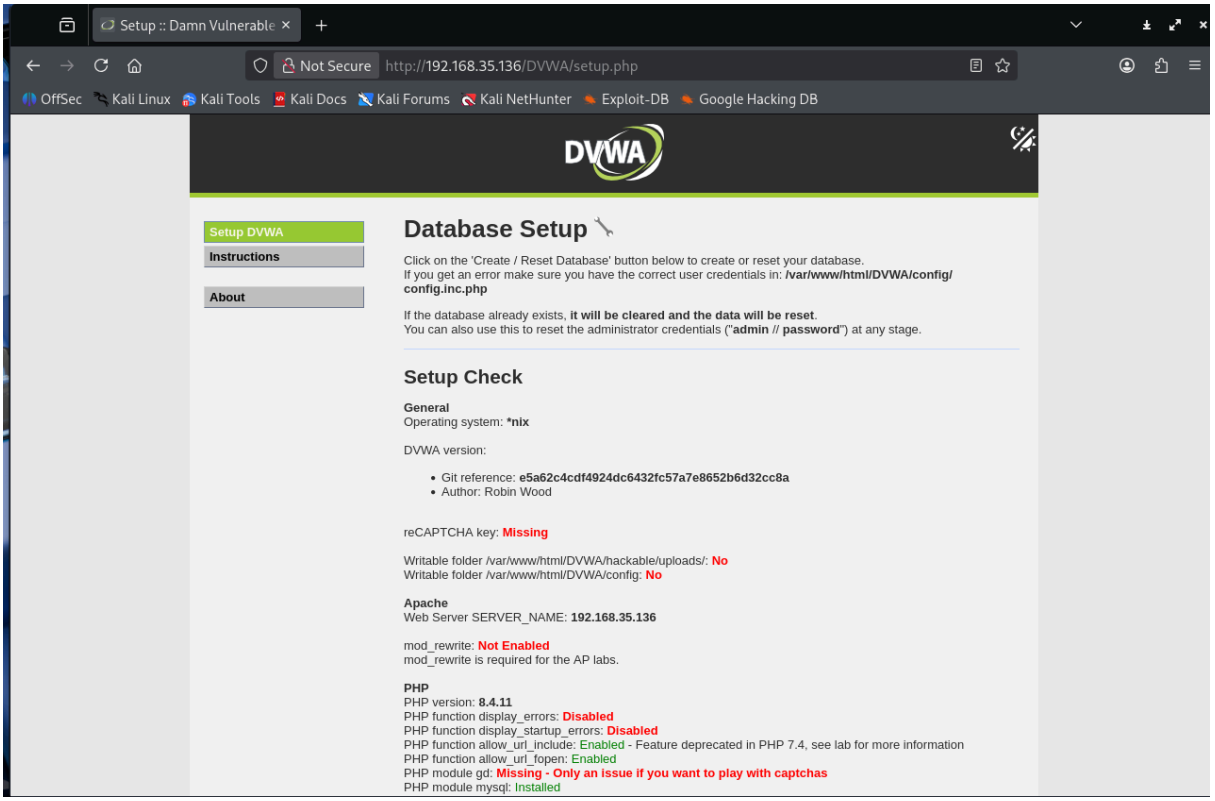
- sudo mysql

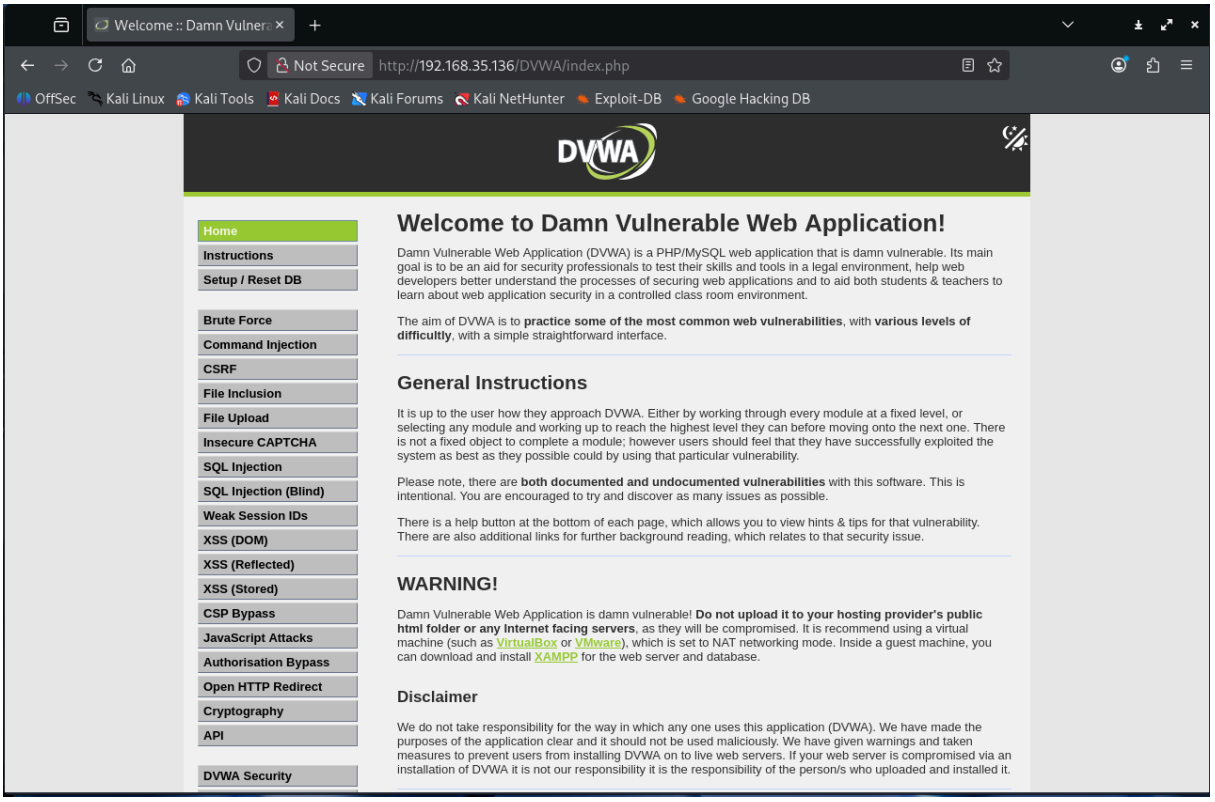
```
┌──(kali㉿kali)-[/var/www/html/DVWA/config]
└─$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 32
Server version: 11.8.3-MariaDB-1+b1 from Debian -- Please help get to 10k stars at https://gi
thub.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
MariaDB [(none)]> CREATE DATABASE dvwa;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> CREATE USER 'dvwa'@'localhost'IDENTIFIED BY'dvwa';
Query OK, 0 rows affected (0.046 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON dvwa. *TO'dvwa'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
Bye

┌──(kali㉿kali)-[/var/www/html/DVWA/config]
└─$ sSsS
```

- sudo nano/etc/php/*/apache2/php.ini
  - allow_url_include = On

```
kali@kali: /var/www/html/DVWA/config
Session  Actions  Edit  View  Help
  GNU nano 8.6                    /etc/php/8.4/apache2/php.ini

; Maximum number of files that can be uploaded via a single request
max_file_uploads = 20

;;;;;;;;;;;;;;;;;;;
; Fopen wrappers ;
;;;;;;;;;;;;;;;;;;;

; Whether to allow the treatment of URLs (like http:// or ftp://) as files.
; https://php.net/allow-url-fopen
allow_url_fopen = On

; Whether to allow include/require to open URLs (like https:// or ftp://) as files.
; https://php.net/allow-url-include
allow_url_include = On

; Define the anonymous ftp password (your email address). PHP's default setting
; for this is empty.
; https://php.net/from
;from="john@doe.com"

; Define the User-Agent string. PHP's default setting for this is empty.
; https://php.net/user-agent
;user_agent="PHP"

; Default timeout for socket based streams (seconds)
; https://php.net/default-socket-timeout
default_socket_timeout = 60


^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

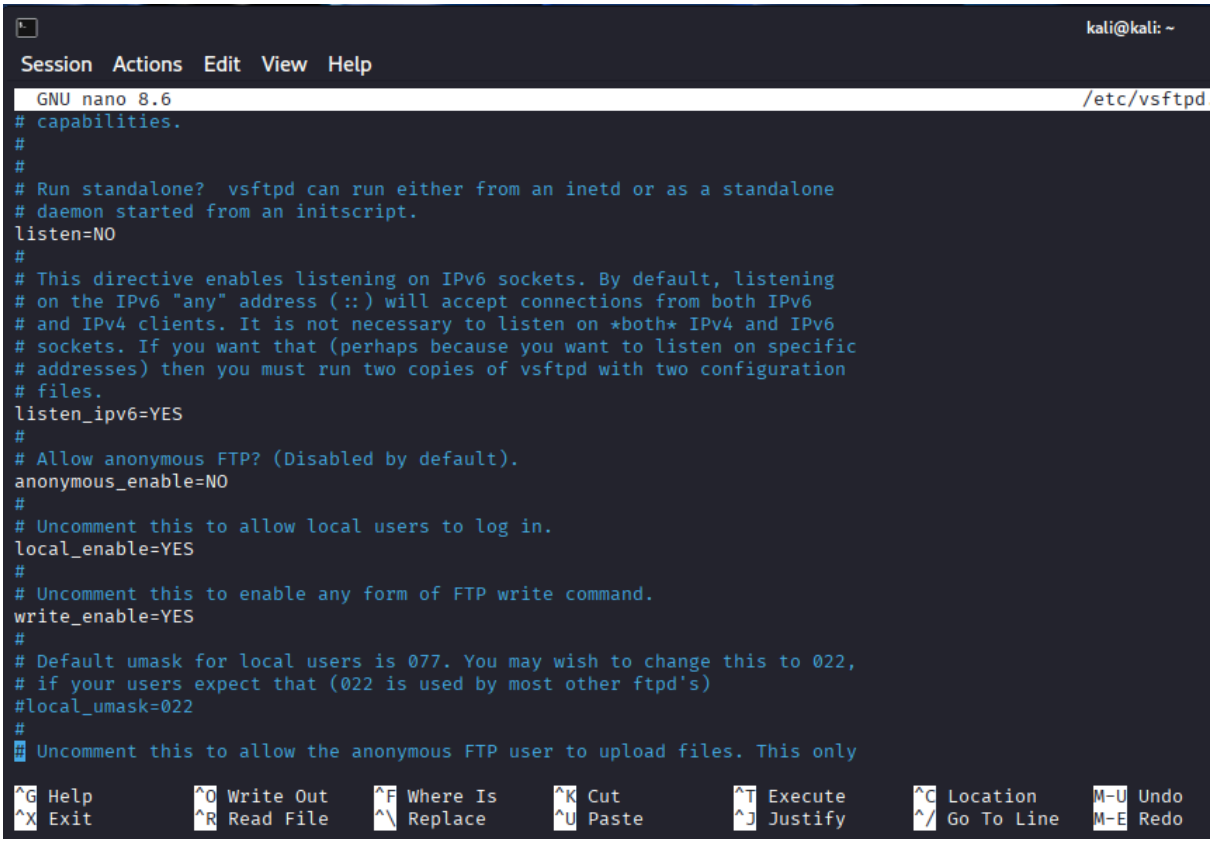
On Client

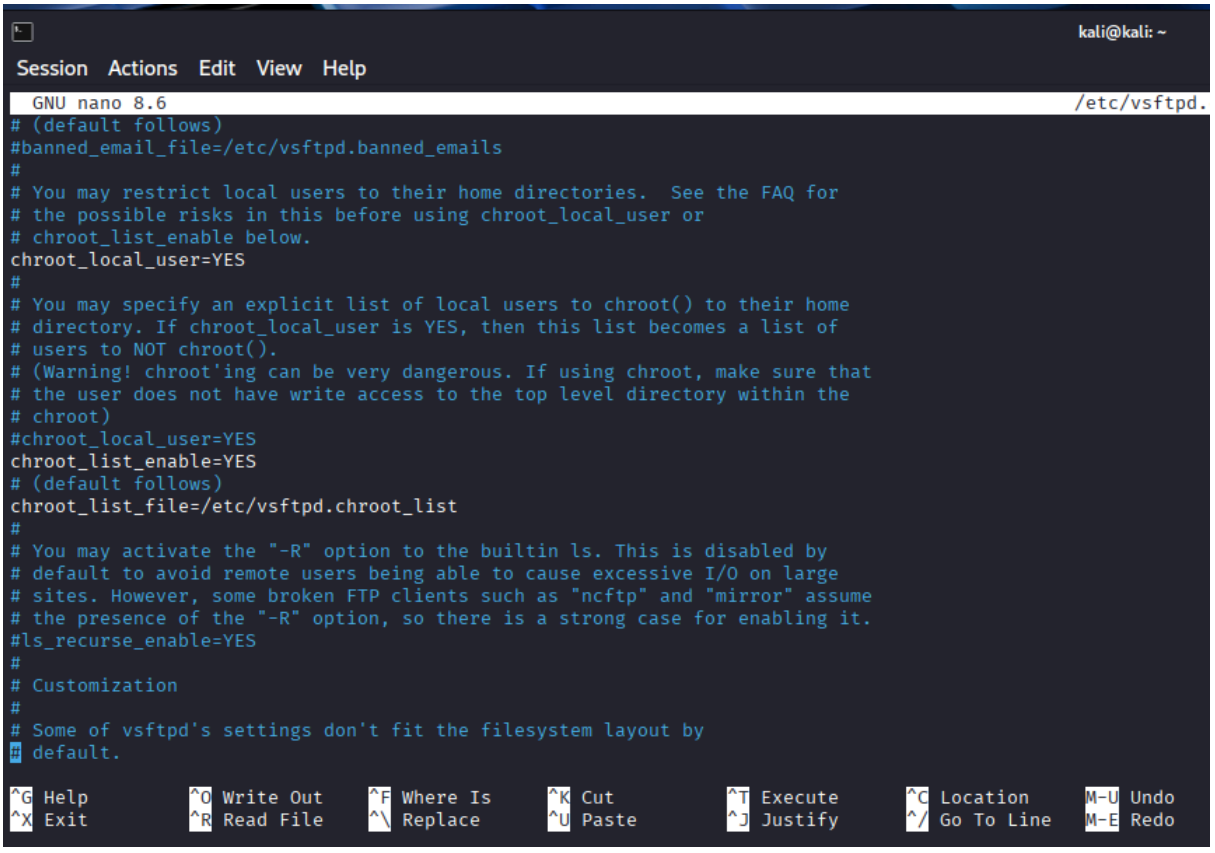




Install and config FTP

- sudo apt install vsftpd -y
- sudo nano /etc/vsftpd.conf





- sudo touch vsftpd.chroot_list
- sudo nano vsftpd.chroot_list

```
kali@kali: ~
Session  Actions  Edit  View  Help
  GNU nano 8.6                    vsftpd.chroot_list *
administator

^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

On Cliernt

```
┌──(kali㉿kali)-[~]
└─$ ftp 192.168.35.136
Connected to 192.168.35.136.
220 (vsFTPd 3.0.5)
Name (192.168.35.136:kali): administrator
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
zsh: suspended  ftp 192.168.35.136
```

Capture packet

On Telnet

On Client

```
┌──(kali㉿kali)-[~]
└─$ telnet 192.168.35.136 23
Trying 192.168.35.136 ...
Connected to 192.168.35.136.
Escape character is '^]'.

Linux 6.12.38+kali-amd64 (kali) (pts/4)

kali login: administrator
Password:
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_6
4

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
┌──(administrator㉿kali)-[~]
└─$
```

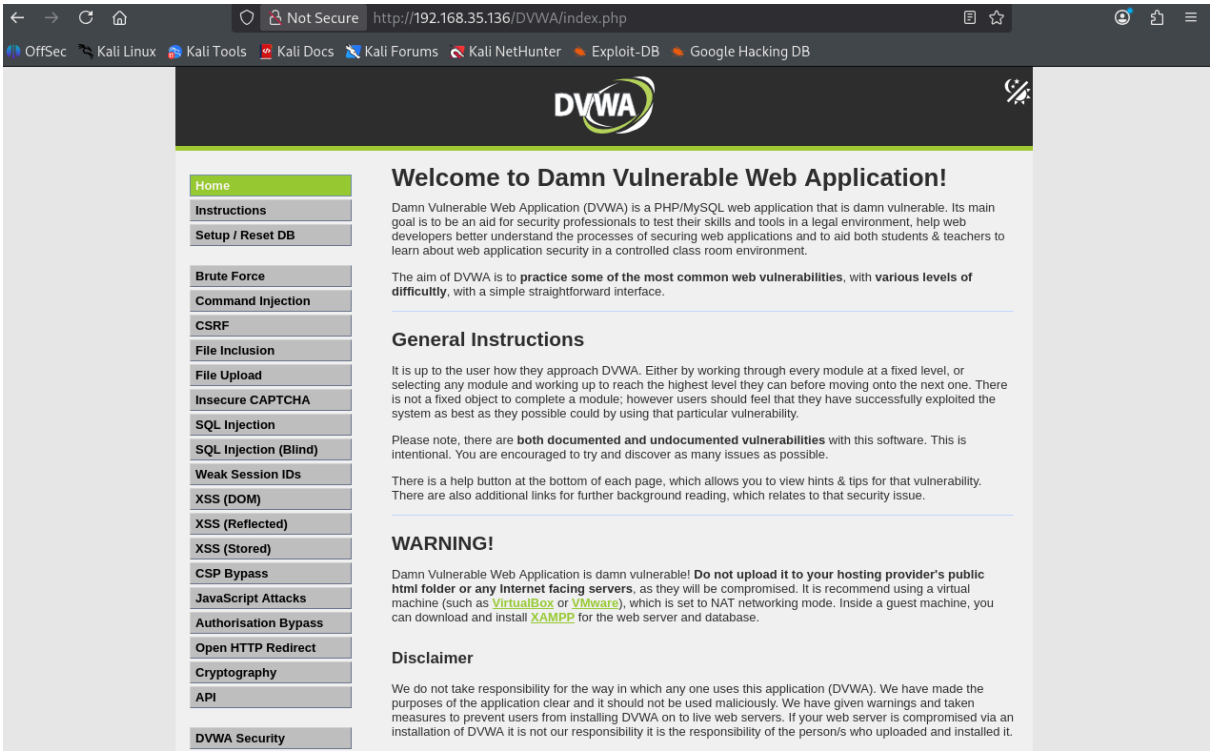
On Attacker

ip.src == 192.168.35.137 && telnet && tcp.len >0

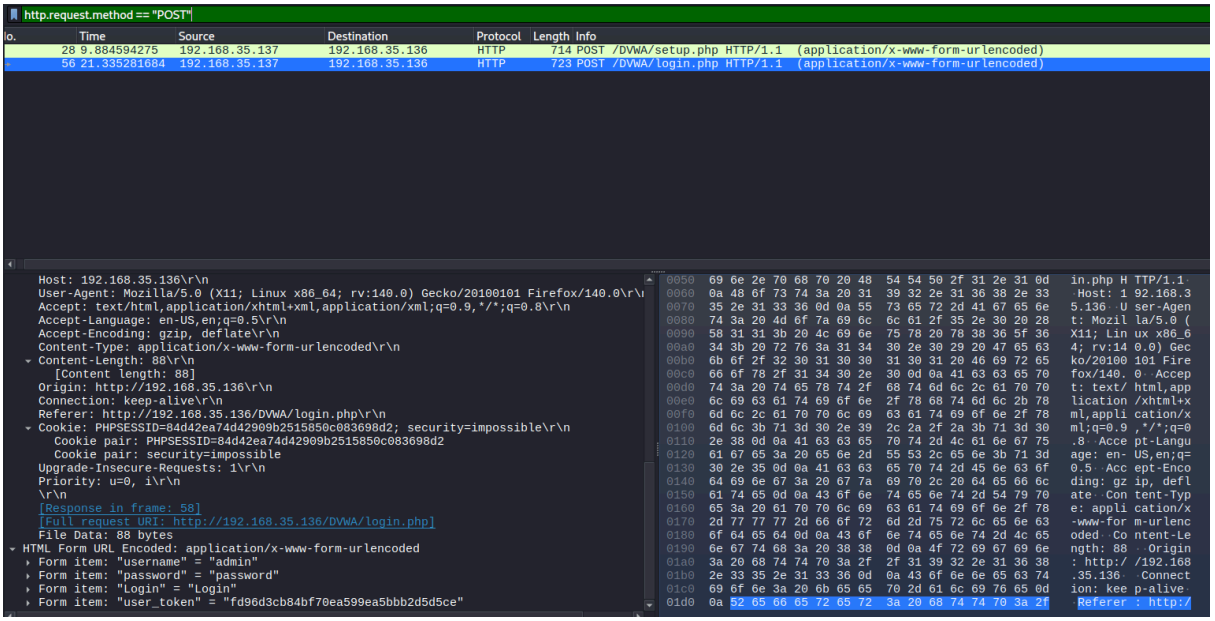
No.	Time	Source	Destination	Protocol	Length	Data	Info
12	1.426931462	192.168.35.137	192.168.35.136	TELNET	93		Do Authentication Option, Do Encryption Option, Suboption Encryption Option, Will Terminal Type
15	1.427393569	192.168.35.137	192.168.35.136	TELNET	140		Suboption Terminal Speed, Suboption X Display Location, Suboption New Environment Option, Subop
17	1.428268744	192.168.35.137	192.168.35.136	TELNET	147		Do Suppress Go Ahead, Won't Echo, Will Linemode, Suboption Linemode, Will Negotiate About Windo
19	1.429144945	192.168.35.137	192.168.35.136	TELNET	73		Suboption Linemode
21	1.429506796	192.168.35.137	192.168.35.136	TELNET	75		Do Echo, Will Binary Transmission, Won't Linemode
37	26.549080515	192.168.35.137	192.168.35.136	TELNET	67	a	1 byte data
40	27.299365304	192.168.35.137	192.168.35.136	TELNET	67	d	1 byte data
43	28.086198464	192.168.35.137	192.168.35.136	TELNET	67	m	1 byte data
46	28.580914816	192.168.35.137	192.168.35.136	TELNET	67	i	1 byte data
49	28.956090073	192.168.35.137	192.168.35.136	TELNET	67	n	1 byte data
52	29.676023502	192.168.35.137	192.168.35.136	TELNET	67	i	1 byte data
55	29.983748566	192.168.35.137	192.168.35.136	TELNET	67	s	1 byte data
58	30.306055231	192.168.35.137	192.168.35.136	TELNET	67	t	1 byte data
61	30.711330968	192.168.35.137	192.168.35.136	TELNET	67	r	1 byte data
64	31.238425314	192.168.35.137	192.168.35.136	TELNET	67	a	1 byte data
67	31.701343261	192.168.35.137	192.168.35.136	TELNET	67	t	1 byte data
70	32.061704836	192.168.35.137	192.168.35.136	TELNET	67	o	1 byte data
73	32.240998176	192.168.35.137	192.168.35.136	TELNET	67	r	1 byte data
76	33.036728250	192.168.35.137	192.168.35.136	TELNET	67	\r	1 byte data
81	34.252314557	192.168.35.137	192.168.35.136	TELNET	67	1	1 byte data
83	34.566041241	192.168.35.137	192.168.35.136	TELNET	67	2	1 byte data
85	34.859296940	192.168.35.137	192.168.35.136	TELNET	67	3	1 byte data
87	35.406230740	192.168.35.137	192.168.35.136	TELNET	67	\r	1 byte data

On HTTP

On Client



On Attacker



On FTP

On Client

```
┌──(kali㉿kali)-[~]
└─$ ftp 192.168.35.136
Connected to 192.168.35.136.
220 (vsFTPd 3.0.5)
Name (192.168.35.136:kali): administrator
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

On Attacker

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ftp

No.	Time	Source	Destination	Protocol	Length	Info
4	0.005895256	192.168.35.136	192.168.35.137	FTP	86	Response: 220 (vsFTPd 3.0.5)
13	14.030655558	192.168.35.137	192.168.35.136	FTP	86	Request: USER administrator
15	14.030949588	192.168.35.136	192.168.35.137	FTP	100	Response: 331 Please specify the password.
23	25.108597984	192.168.35.137	192.168.35.136	FTP	76	Request: PASS 123
25	25.205197214	192.168.35.136	192.168.35.137	FTP	89	Response: 230 Login successful.
27	25.205688549	192.168.35.137	192.168.35.136	FTP	72	Request: SYST
29	25.205964732	192.168.35.136	192.168.35.137	FTP	85	Response: 215 UNIX Type: L8
30	25.206776363	192.168.35.137	192.168.35.136	FTP	72	Request: FEAT
31	25.207223741	192.168.35.136	192.168.35.137	FTP	81	Response: 211-Features:
32	25.207223850	192.168.35.136	192.168.35.137	FTP	87	Response: EPRT
33	25.207223874	192.168.35.136	192.168.35.137	FTP	110	Response: PASV